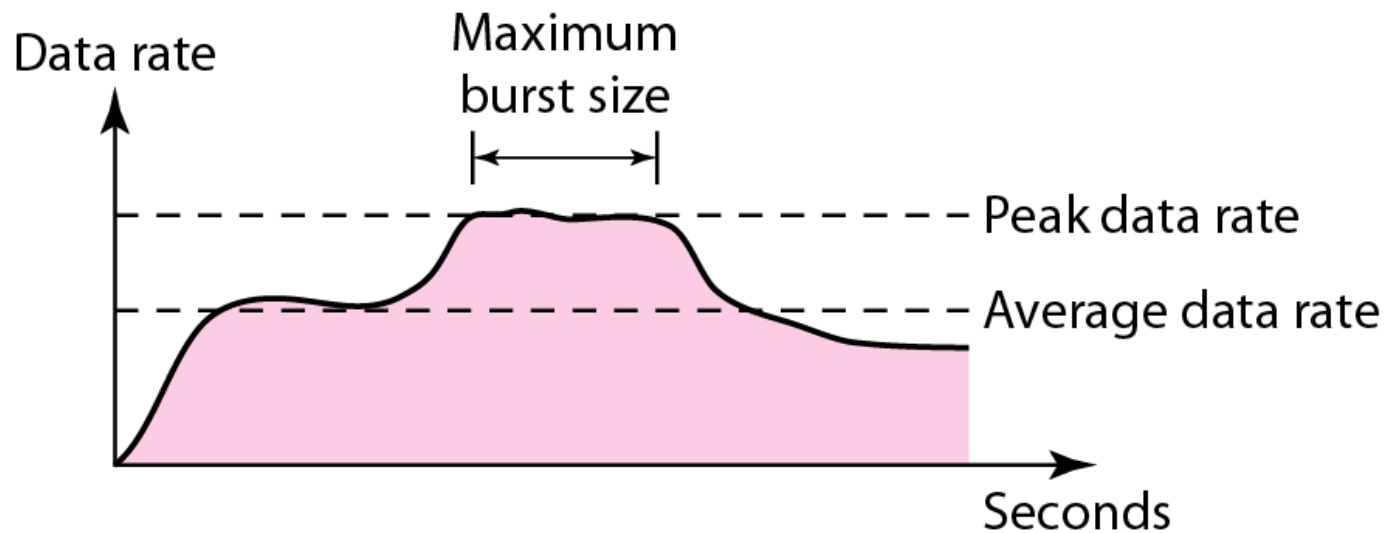


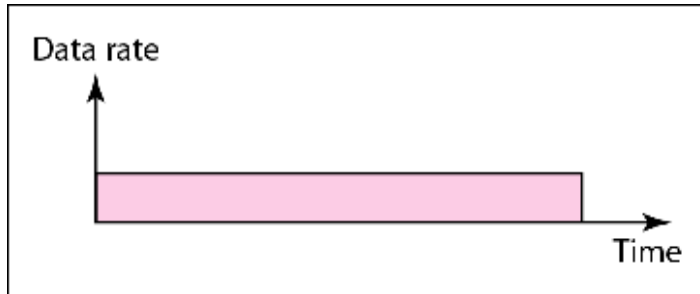
DATA TRAFFIC

- Focus: QoS and data traffic
- Congestion control avoids traffic congestion
- QoS ensures a proper traffic environment
- QoS improves user experience

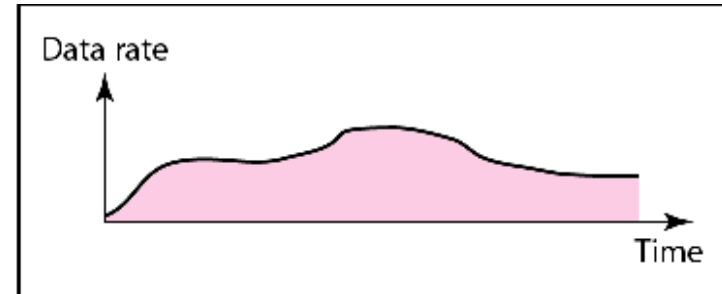
Traffic descriptors



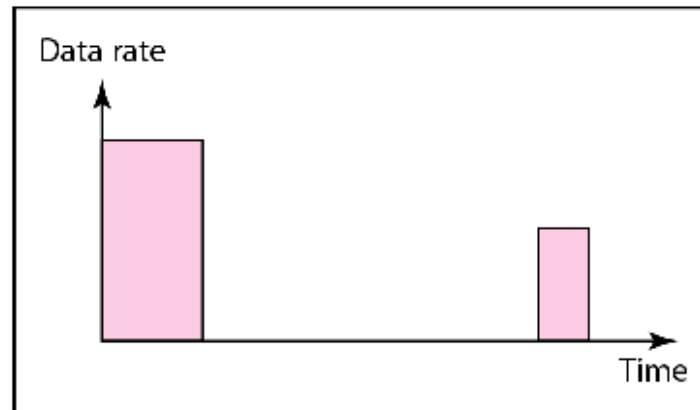
Three traffic profiles



a. Constant bit rate



b. Variable bit rate



c. Bursty

CONGESTION

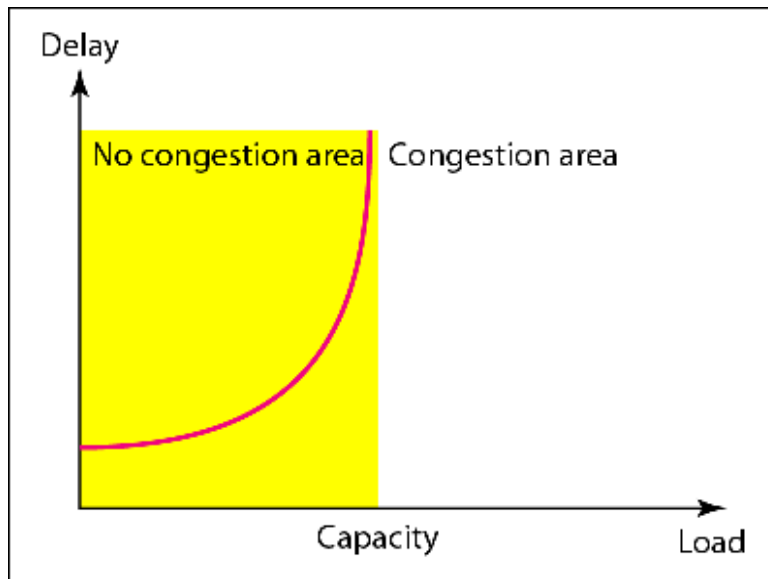
Network Congestion

- Congestion occurs when $\text{load} > \text{capacity}$
- Load = packets sent to network
- Capacity = packets network can handle
- Congestion control uses mechanisms, techniques
- Goal: keep load below capacity

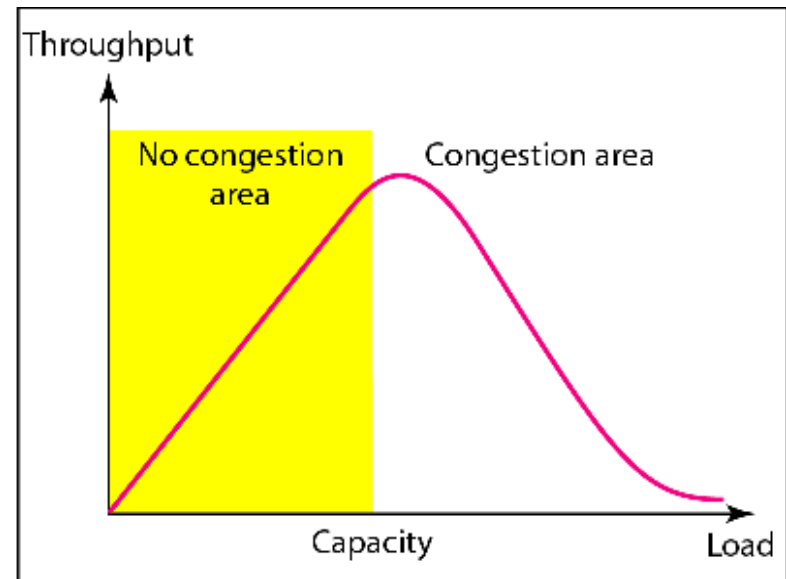
Effects of Congestion Control

- Improved Network Stability: Prevents overload and keeps the network running smoothly.
- Reduced Latency and Packet Loss: Ensures data is delivered faster with fewer retransmissions.
- Enhanced Throughput: Maximizes the volume of data successfully transferred in a given time.
- Fairness in Resource Allocation: Distributes bandwidth evenly so no user monopolizes resources.
- Better User Experience: Provides faster access to websites, applications, and services.
- Prevention of Congestion Collapse: Avoids severe breakdowns where the network becomes nearly unusable.

Network Performance: Delay and throughput as functions of load



a. Delay as a function of load

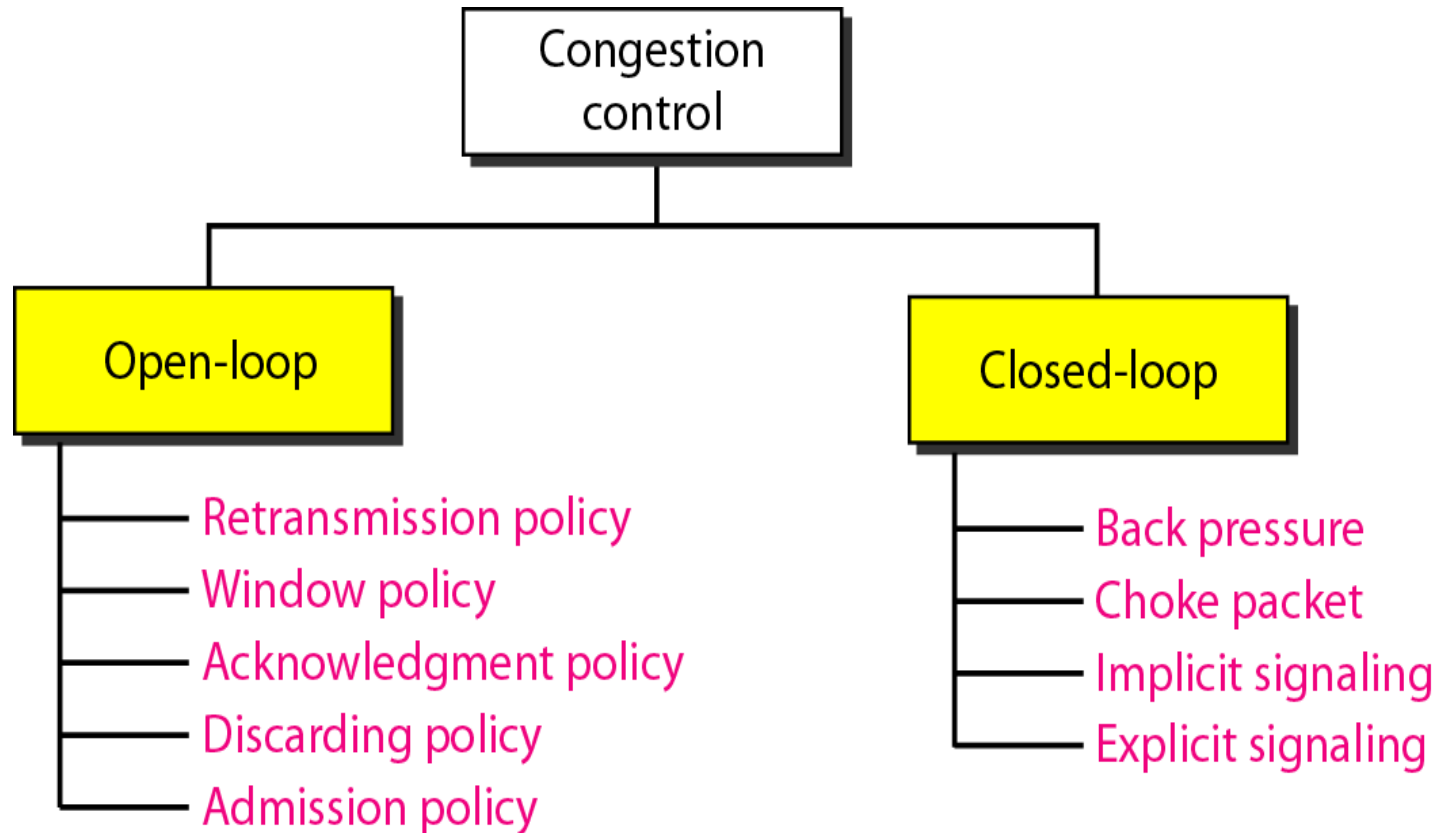


b. Throughput as a function of load

CONGESTION CONTROL

- Uses techniques and mechanisms
- Prevent congestion before it happens
- Remove congestion after it occurs
- Two main categories exist
- Open- loop: prevention methods
- Closed- loop: removal methods

Congestion control categories



Retransmission Policy

- Retransmit on lost frame
- Retransmit on lost ack
- Retransmit on damaged frame
- After timer expiration
- Small timer more retransmits
- Large timer delayed response

Window policy

- Defines maximum window size
- Used in sliding window protocol
- Two possible implementations
- Go- Back N ARQ
- Selective Repeat ARQ

Acknowledgement Policy

- Selective Ack
- Cumulative Ack

Discarding policy

- Applied during network congestion
- Decide which packet to drop
- Drop low priority packets first
- Drop newer packets if needed
- Avoid dropping near- destination packets
- Ensure fairness in discarding

Admission Policy

- Applied when resources are low
- Do not accept extra packets
- Prevent overload from sender
- Router may deny connection
- Deny new virtual circuits
- Helps avoid further congestion

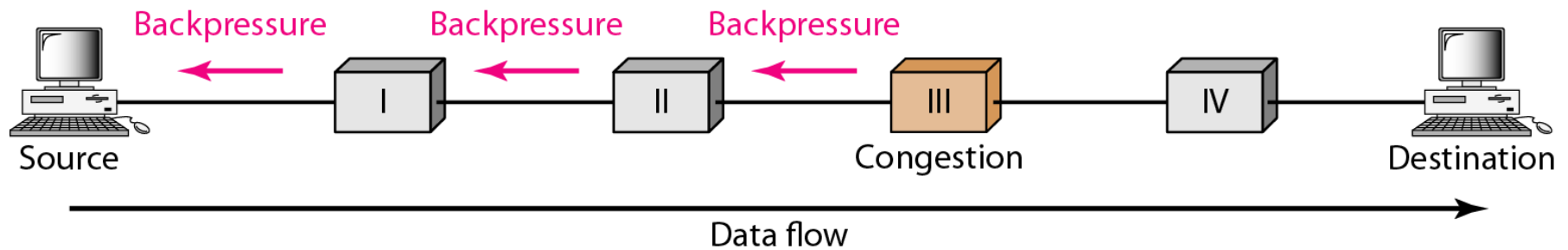
Solutions to Open Loop Congestion

- Use selective repeat window
- Prefer cumulative acknowledgments
- Avoid discarding high- priority packets

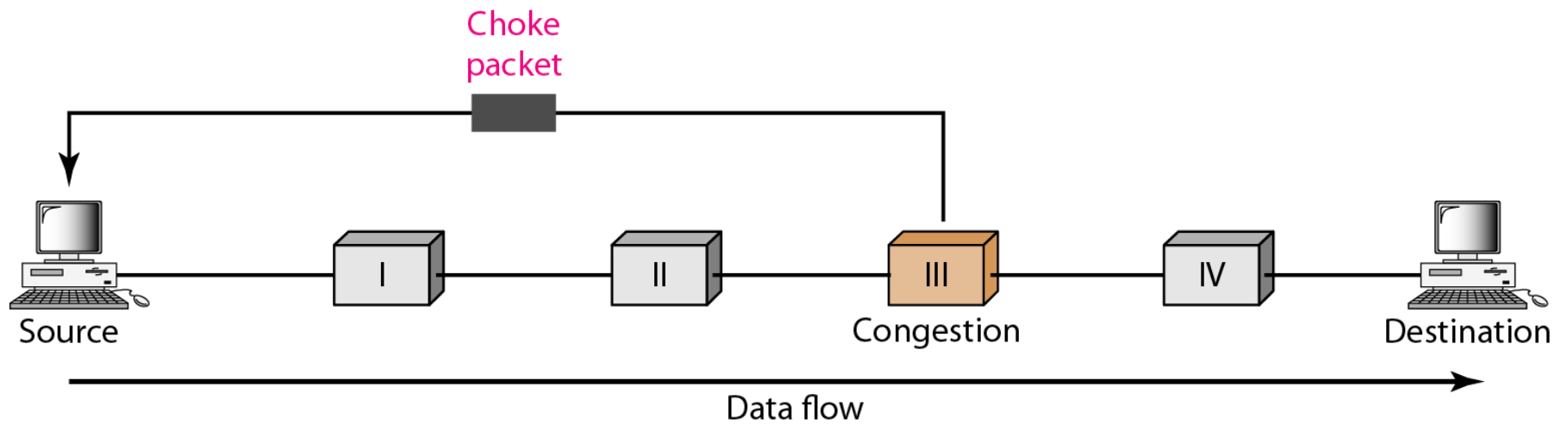
Closed Loop Congestion Control

- Backpressure
- Choke packet
- Implicit Signaling
- Explicit Signaling

Backpressure method for alleviating congestion



Choke packet



Closed Loop Congestion Control

- Implicit: Source infers by symptoms
- Example: delayed acknowledgments
- Explicit: Network sends direct alert
- Different from choke packet
- Backward: Warns the source, take measures
- Forward: Warns destination, slow ACKs

Congestion leads to –

- discarding of segments
- transmission delay increase

Quality of service (QoS)

- Provide network with enough capacity
- Called solution: Over- provisioning
- Handles all possible traffic load
- QoS helps with limited capacity
- Meets application requirements
- Achieves performance at lower cost

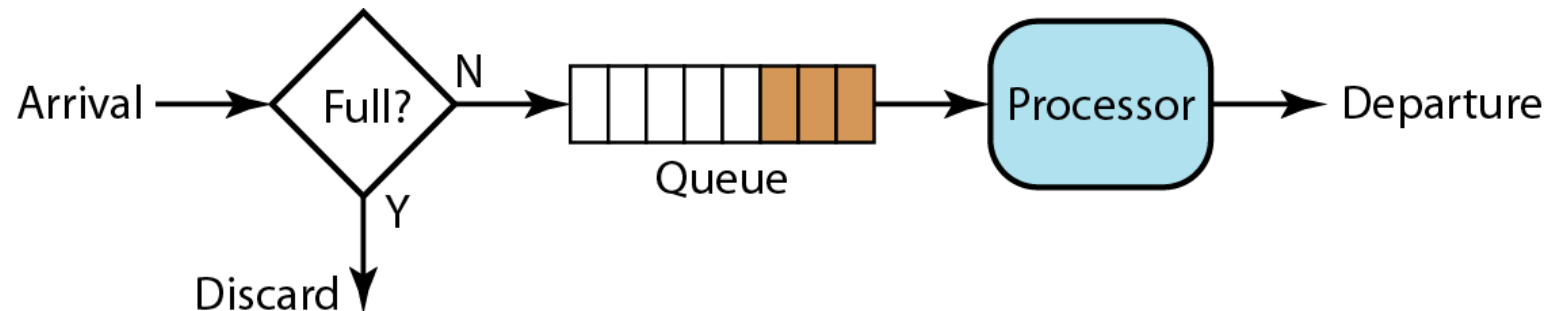
Flow Characteristics

- Reliability
- Delay
- Jitter – variation in delay
- Bandwidth

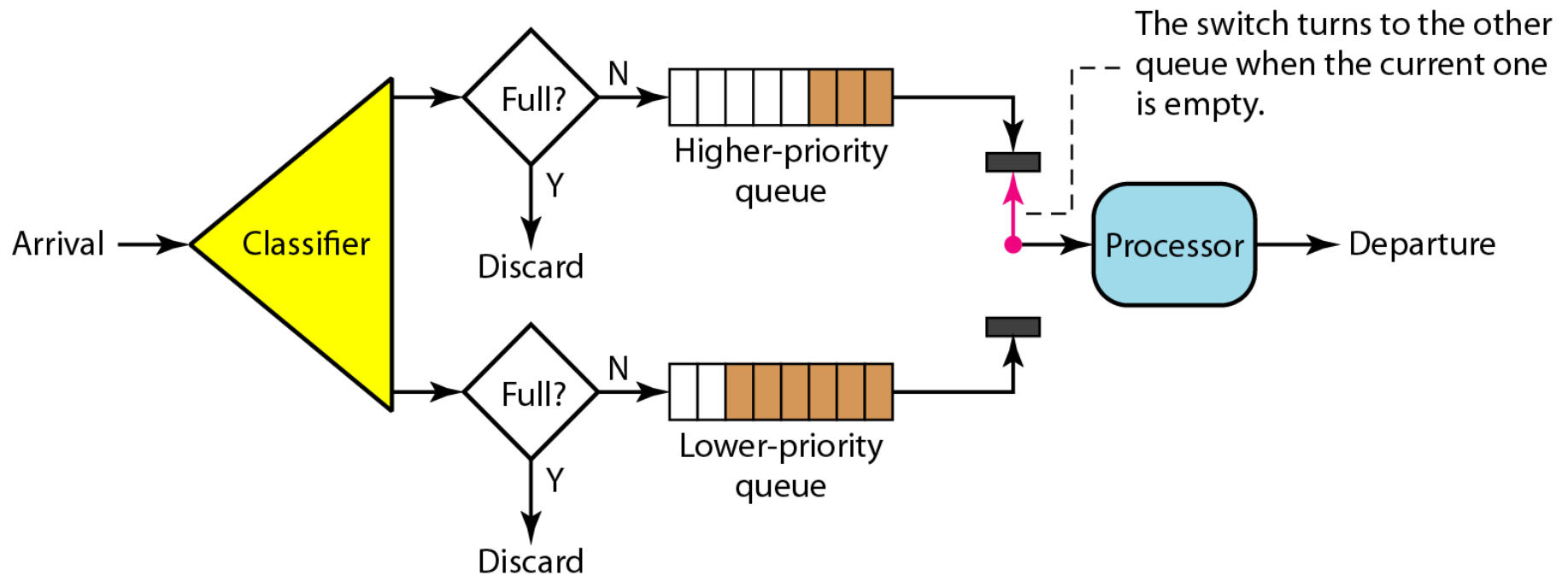
TECHNIQUES TO IMPROVE QoS

- Scheduling
 - FIFO Queuing
 - Priority Queuing
 - Weighted Fair Queuing
- Traffic Shaping
 - Leaky Bucket
 - Token Bucket
- Resource Reservation
- Admission Control

Scheduling : FIFO queue

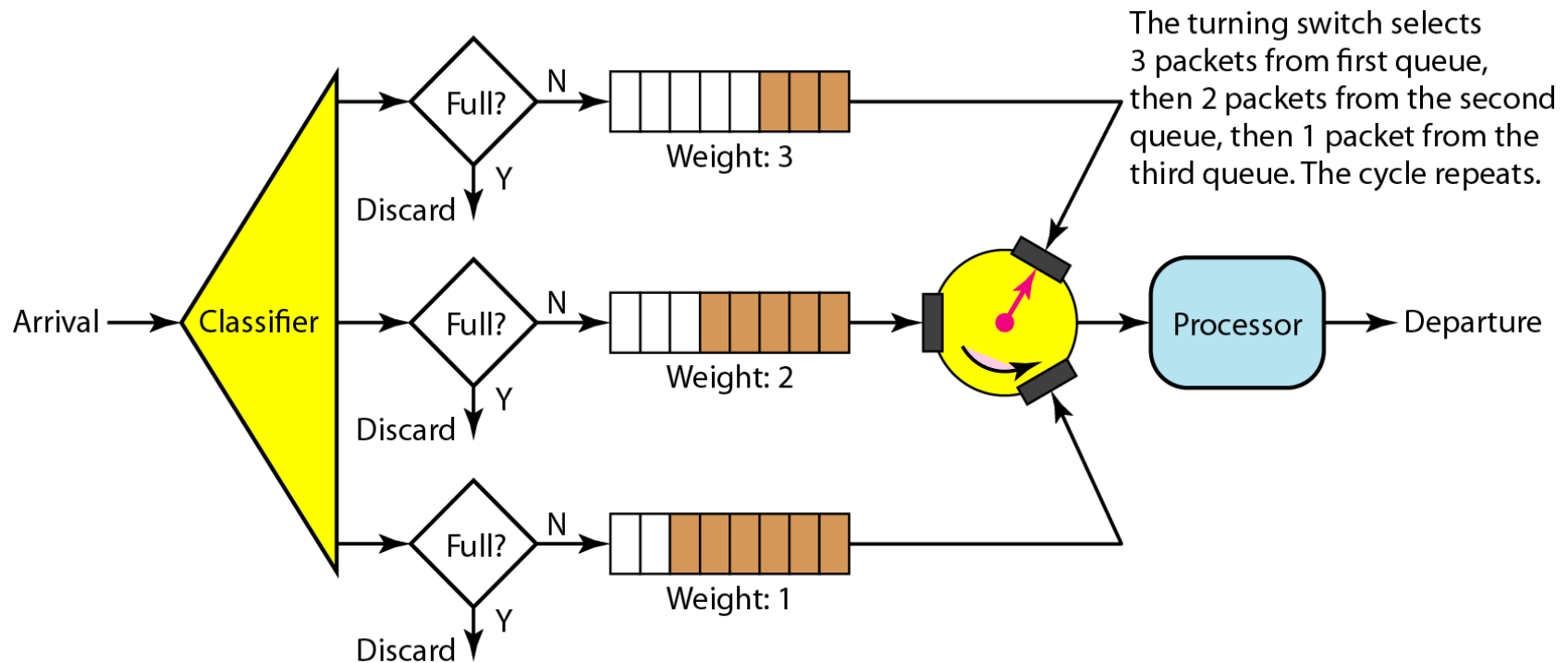


Scheduling : Priority queuing



- Priority queuing have problem of starvation

Scheduling : Weighted fair queuing



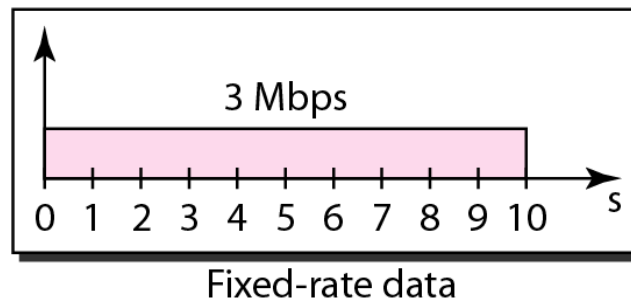
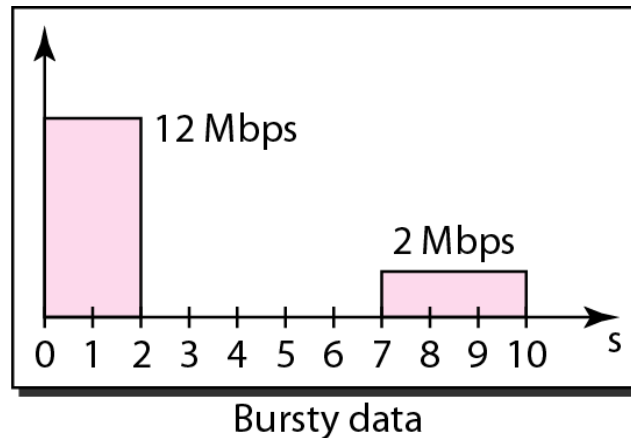
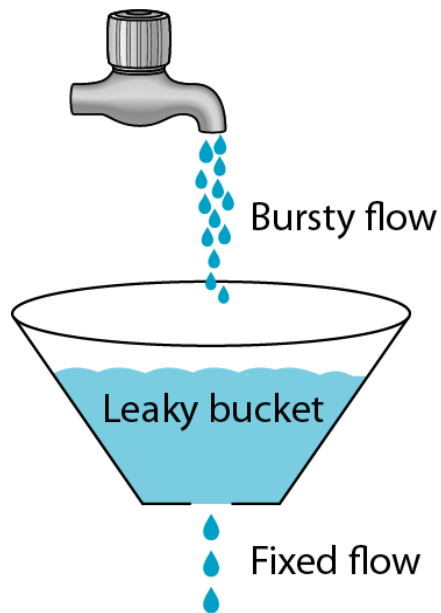
Traffic Shaping

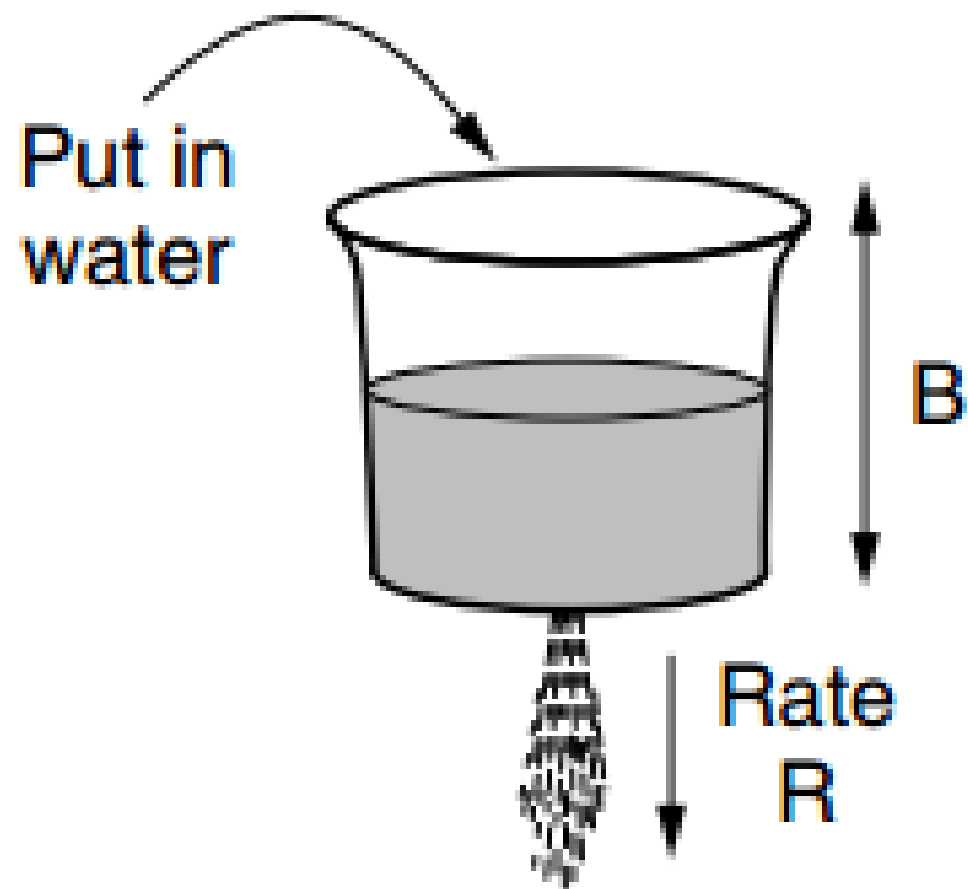
- Regulates average data flow rate
- Controls the burstiness of traffic
- Allows flexible application needs
- Supports occasional traffic bursts
- Ensures fairness in transmission
- Related to Service Level Agreement (SLA)

Traffic Policing

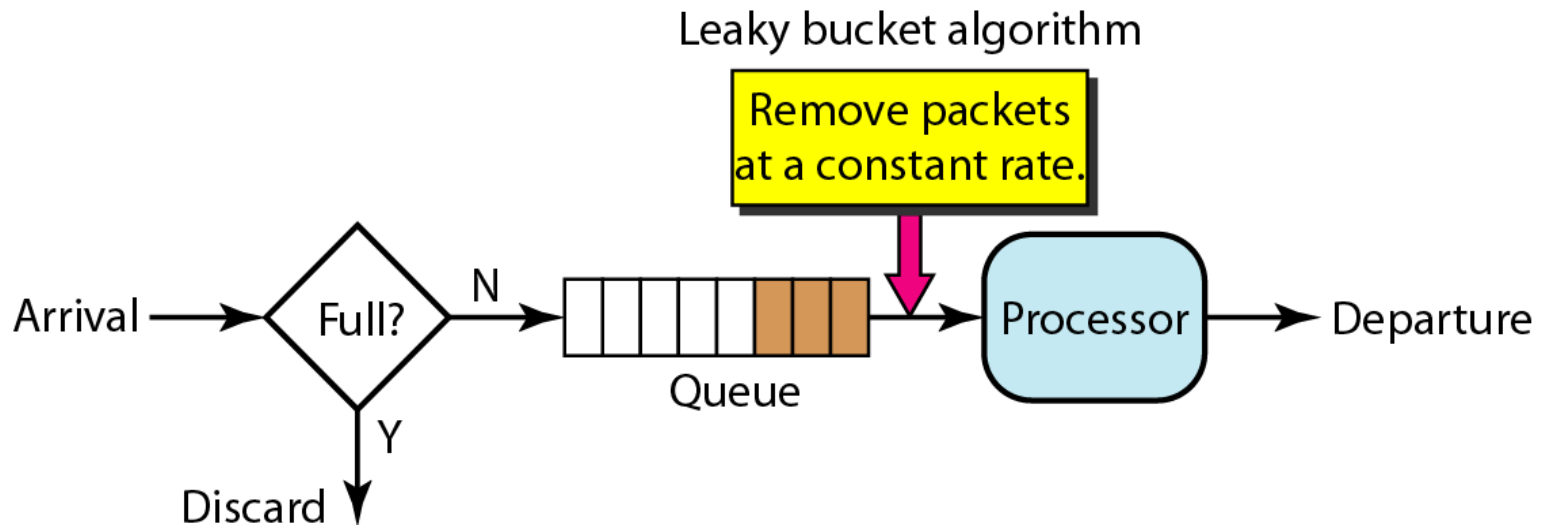
- Traffic shaping reduces congestion
- Ensures network meets performance promise
- Provider monitors user compliance
- Excess packets may be dropped
- Or marked as lower priority
- Monitoring process called traffic policing

Traffic Shaping : Leaky bucket





Traffic Shaping : Leaky bucket implementation

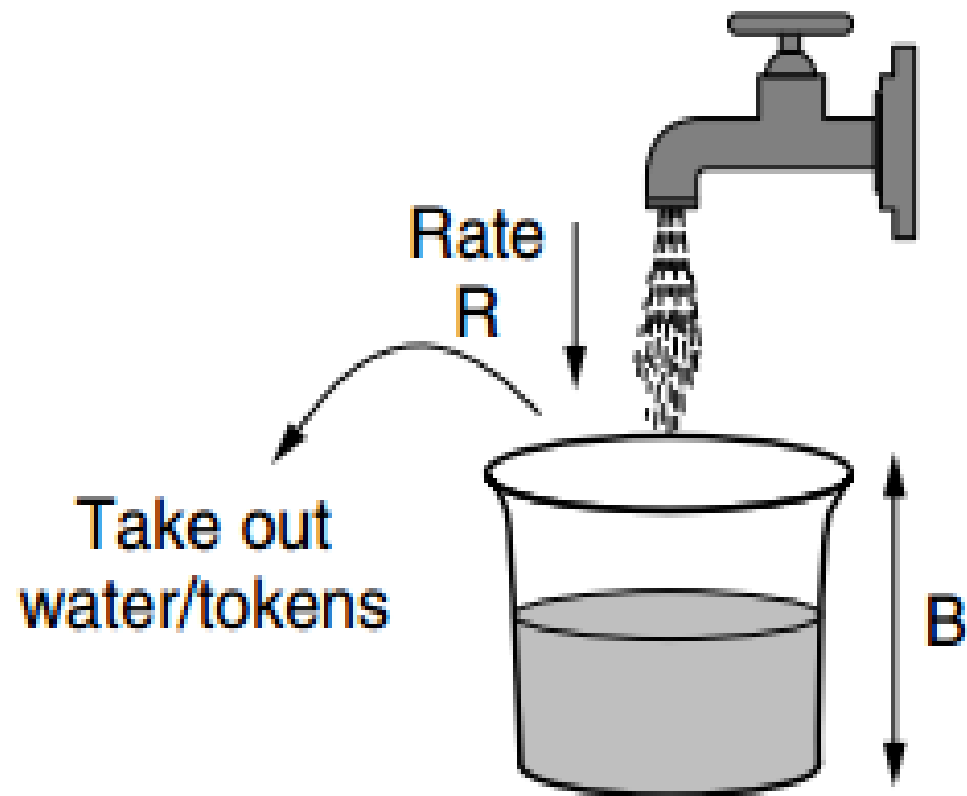


Note

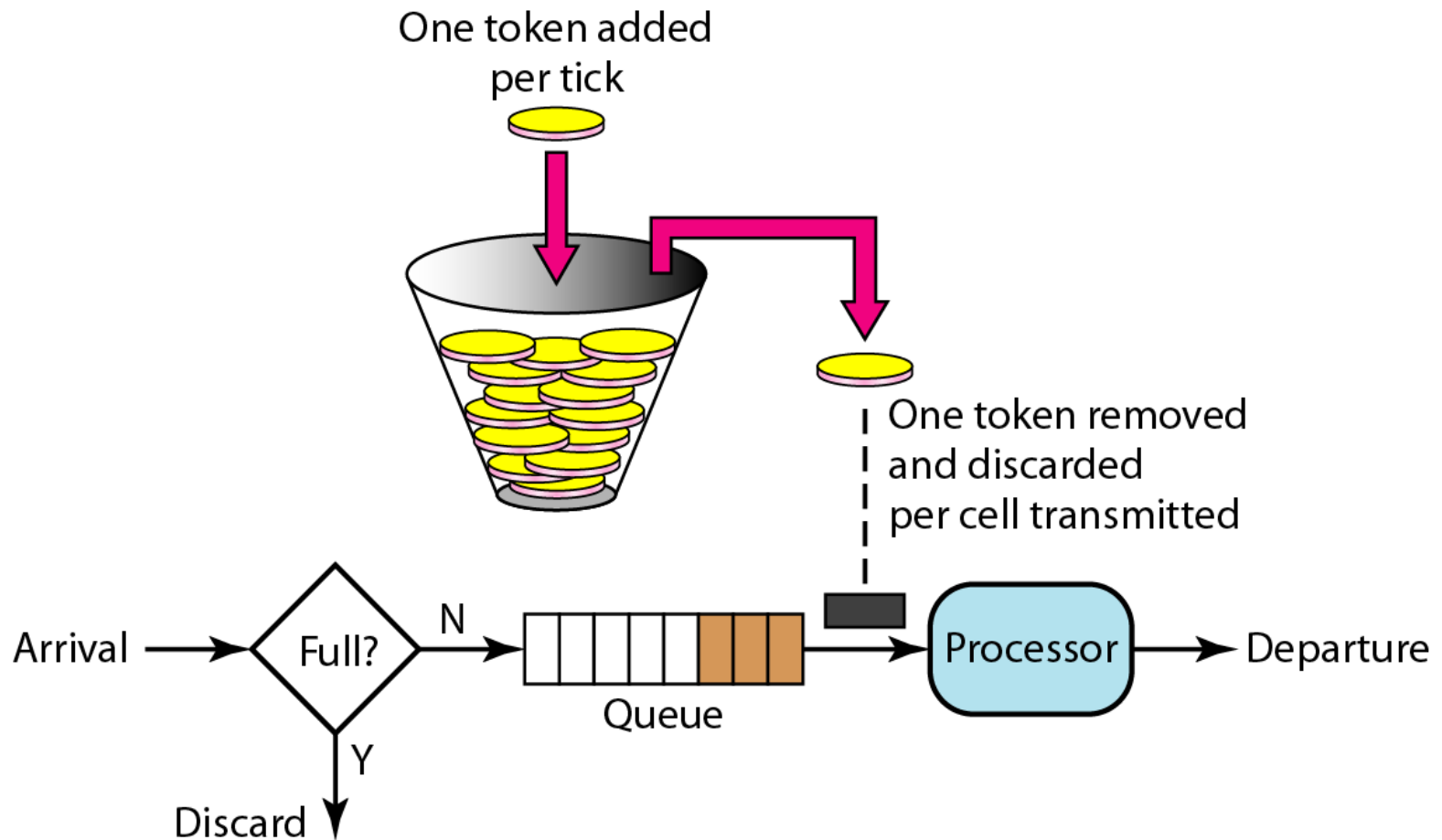
A leaky bucket algorithm shapes bursty traffic into fixed- rate traffic by averaging the data rate. It may drop the packets if the bucket is full.

Note

The token bucket allows bursty traffic at a regulated maximum rate.



Token bucket



Puzzle

Puzzle 1: Match the Algorithm

Match the congestion control algorithm with its key feature:

Algorithm	Features (Randomized Order)
Leaky Bucket	(A) Packets sent at fixed rate
Token Bucket	(B) Tokens control transmission burst
TCP Tahoe	(C) Uses Slow Start + AIMD
TCP Reno	(D) Fast Retransmit + Fast Recovery